

Title: 'Peer-graded Assignment: "Statistical Inference Course Project"'

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Part 2: Instructions for Basic Inferential Data Analysis

In the second part of the project, we'll look at the ToothGrowth data in the R datasets package.

Load the ToothGrowth data and conduct some exploratory data analysis.

Give a brief summary of the data.

To compare tooth growth by supplement and dose, use confidence intervals and/or hypothesis tests. (Even if other approaches are worth considering, only use the techniques from class.)

State your conclusions as well as the assumptions that led to them.

Part 1 of the Resolution

To begin, we run 1000 simulations of 40 exponentials.

Required libraries

```
library(ggplot2)
```

```
library(knitr)
```

setting seed

```
set.seed(1)
```

#No. of values (n) = 40, lambda = 0.2, No.of iterations, at least 1000, numsim=2000, Theoretical mean = $1/\lambda$ or $1/0.2$

```
lambda <- 0.2
```

```
nosim <- 1:1000 # Number of Simulations/rows
```

```
n <- 40
```

Using rexp to generate data

Create a dataset with the mean and lambda values specified above by using the rexp function.

```
#sd(apply(matrix(rnorm(nosim*n), nosim), 1, mean))
```

#Create a matrix of simulated values:

```
e_matrix <- data.frame(x = sapply(nosim, function(x) {mean(rexp(n, lambda))}))
head(e_matrix)
```

1. Display the distribution's center and compare it to the theoretical distribution's center.

```
sim_mean <- apply(e_matrix, 2, mean)
sim_mean
```

Which is very close to the expected theoretical center of the distribution:

```
th_mean <- 1/lambda
Th_mean
```

2.Explain how variable it is and compare it to the distribution's theoretical variance.

```
sim_SD <- sd((e_matrix$x))
sim_SD
sim_Var <- var(e_matrix$x)
sim_Var
```

Let's compare, the expected theoretical SD and Variance are:

```
th_SD <- (1/lambda)/sqrt(n)
th_SD
```

```
th_Var <- th_SD^2
th_Var
```

Table Comparing Theoretical and Actual Mean, Standard Deviation, and Variance Values

Actual Val Variable Theoretical Val

Mean 5 5.048

SD 0.791 0.796

Var 0.625 0.634

We can confirm that the differences are, as expected, minimal.

3. Demonstrate that the distribution is roughly normal. We conclude that the function appears to approximate nearly normal.

```
plot <- ggplot(data = e_matrix, aes(x = x)) +  
  geom_histogram(aes(y=..density..), binwidth = 0.20, fill="slategray3", col="black")  
plot <- plot + labs(title="Density of 40 Numbers from Exponential Distribution", x="Mean of 40 Selections",  
y="Density")  
plot <- plot + geom_vline(xintercept=sim_mean,size=1.0, color="black")  
plot <- plot + stat_function(fun=dnorm,args=list(mean=sim_mean, sd=sim_SD),color = "dodgerblue4",  
size = 1.0)  
plot <- plot+ geom_vline(xintercept=th_mean,size=1.0,color="indianred4",linetype = "longdash")  
plot <- plot + stat_function(fun=dnorm,args=list(mean=th_mean, sd=th_SD),color = "darkmagenta", size  
= 1.0)  
plot
```

Resolution Part 2

1. We load the ToothGrowth data in the R datasets package and perform some basic exploratory data analyses

```
library(datasets)  
data(ToothGrowth)  
head(ToothGrowth)
```

EDA

The Effect of Vitamin C on Guinea Pig Tooth Growth

Description:

The length of odontoblasts (cells responsible for tooth growth) in 60 guinea pigs is the response. Each animal was given one of three vitamin C dose levels (0.5, 1, and 2 mg/day) via one of two delivery methods: orange juice or ascorbic acid (a form of vitamin C and coded as VC).

Format

A data frame containing 60 observations on three variables.

[,1] numeric tooth length

[,2] supplement factor (VC or OJ)

Dose in milligrams/day [,3]

We can easily visualize the dataset using a boxplot and a coplot:

```
boxplot(len ~ supp * dose, data=ToothGrowth, ylab="Tooth Length", main="Comparing Tooth Growth  
between different supplements and different dosis", col=c("brown4", "cornflowerblue", "brown4",  
"cornflowerblue", "brown4", "cornflowerblue"))
```

```
require(graphics)
```

```
coplot(len ~ dose | supp, data = ToothGrowth, panel = panel.smooth, xlab = "Comparing Tooth Growth  
between different supplements and different dosis", col=c("royalblue4"), pch=21)
```

2. Give a brief summary of the data.

As shown in the plots, the average tooth length appears to increase with supplement dosis. This could imply that there is a link between using a supplement dosis and tooth growth.

Additional data summaries:

```
# basic info
```

```
head(ToothGrowth)
```

```
nrow(ToothGrowth)
```

```
summary(ToothGrowth)
```

```
table(ToothGrowth$supp, ToothGrowth$dose)
```

3. To compare tooth growth by supplement and dose, use confidence intervals and/or hypothesis tests. (Even if other approaches are worth considering, only use the techniques from class.)

We recommend the T distribution because the sample size is not too large.

```
t.test(len ~ supp, data = ToothGrowth)
```

When the difference between the two supplements is compared, there is no strong evidence to reject the null hypothesis because the p-value is greater than the 5% significance level.

However, for this dataset, it is also important to compare the differences between the different dosis levels, because higher dosis levels may produce contradictory evidence.

4. Based on the existing datasets and T statistics, there is no convincing evidence that there is a difference between the two types of supplements, as previously stated. Consequently, we fail to reject the Null hypothesis (H_0).

However, there is compelling evidence that there is a link between dosis level and growth. In other words, we reject the Null hypothesis (H_0) in favor of the alternative hypothesis (H_a).

```
library(ggplot2)
> library(knitr)
>
> set.seed(1)
>
> lambda <- 0.2
> nosim <- 1:1000 # Number of Simulations/rows
> n <- 40
>
> #sd(apply(matrix(rnorm(nosim*n), nosim), 1, mean))
> #Create a matrix of simulated values:
> e_matrix <- data.frame(x = sapply(nosim, function(x) {mean(rexp(n, lambda))})))
```

```

> head(e_matrix)

      x
1 4.860372
2 5.961285
3 4.279204
4 4.702298
5 5.196446
6 4.397114

> sim_mean <- apply(e_matrix, 2, mean)

> sim_mean

      x
4.990025

>

> th_mean <- 1/lambda

> th_mean

[1] 5

>

> sim_SD <- sd((e_matrix$x))

> sim_SD

[1] 0.7817394

>

> sim_Var <- var(e_matrix$x)

> sim_Var

[1] 0.6111165

>

> th_SD <- (1/lambda)/sqrt(n)

> th_SD

[1] 0.7905694

>

> th_Var <- th_SD^2

```

```

> th_Var
[1] 0.625
>
> plot <- ggplot(data = e_matrix, aes(x = x)) +
+   geom_histogram(aes(y=..density..), binwidth = 0.20, fill="slategray3", col="black")
> plot <- plot + labs(title="Density of 40 Numbers from Exponential Distribution", x="Mean of 40
Selections", y="Density")
> plot <- plot + geom_vline(xintercept=sim_mean,size=1.0, color="black")
> plot <- plot + stat_function(fun=dnorm,args=list(mean=sim_mean, sd=sim_SD),color = "dodgerblue4",
size = 1.0)
> plot <- plot+ geom_vline(xintercept=th_mean,size=1.0,color="indianred4",linetype = "longdash")
> plot <- plot + stat_function(fun=dnorm,args=list(mean=th_mean, sd=th_SD),color = "darkmagenta", size
= 1.0)
> plot
>
> library(datasets)
> data(ToothGrowth)
> head(ToothGrowth)
  len supp dose
1  4.2   VC  0.5
2 11.5   VC  0.5
3  7.3   VC  0.5
4  5.8   VC  0.5
5  6.4   VC  0.5
6 10.0   VC  0.5
>
> boxplot(len ~ supp * dose, data=ToothGrowth, ylab="Tooth Length", main="Comparing Tooth Growth
between different supplements and different dosis", col=c("brown4", "cornflowerblue", "brown4",
"cornflowerblue", "brown4", "cornflowerblue"))
>
> require(graphics)

```

```
> coplot(len ~ dose | supp, data = ToothGrowth, panel = panel.smooth,xlab = "Comparing Tooth Growth  
between different supplements and different dosis", col=c("royalblue4"), pch=21)
```

```
> # basic info
```

```
> head(ToothGrowth)
```

```
  len supp dose  
1  4.2  VC 0.5  
2 11.5  VC 0.5  
3  7.3  VC 0.5  
4  5.8  VC 0.5  
5  6.4  VC 0.5  
6 10.0  VC 0.5
```

```
>
```

```
> nrow(ToothGrowth)
```

```
[1] 60
```

```
>
```

```
> summary(ToothGrowth)
```

```
  len      supp      dose  
Min.   :4.20  OJ:30  Min.   :0.500  
1st Qu.:13.07 VC:30  1st Qu.:0.500  
Median :19.25      Median :1.000  
Mean   :18.81      Mean    :1.167  
3rd Qu.:25.27      3rd Qu.:2.000  
Max.   :33.90      Max.    :2.000
```

```
>
```

```
> table(ToothGrowth$supp,ToothGrowth$dose)
```

```
  0.5  1  2  
OJ  10 10 10  
VC  10 10 10
```

```
>
```



```
> t.test(len ~ supp, data = ToothGrowth)
```

Welch Two Sample t-test

data: len by supp

t = 1.9153, df = 55.309, p-value = 0.06063

alternative hypothesis: true difference in means is not equal to 0

95 percent confidence interval:

-0.1710156 7.5710156

sample estimates:

mean in group OJ mean in group VC

20.66333 16.96333

```
> t.test(ToothGrowth$len, ToothGrowth$dose)
```

Welch Two Sample t-test

data: ToothGrowth\$len and ToothGrowth\$dose

t = 17.81, df = 59.798, p-value < 2.2e-16

alternative hypothesis: true difference in means is not equal to 0

95 percent confidence interval:

15.66453 19.62881

sample estimates:

mean of x mean of y

18.813333 1.166667

